Pooya Barazandeh

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Summary

Computer Science graduate ranked **1st out of 72 students** at Amirkabir University of Technology with a GPA of **19.55/20**, specializing in AI/ML, federated learning, LLM/RAG systems, and computer vision. Authored a submitted paper on communication-efficient vertical federated learning and completed Oxford Machine Learning School's **MLx Representation Learning & Generative AI 2026** programme. Teaching Assistant for **10+ undergraduate and graduate courses**, including Head TA for Advanced AI, with industry experience leading production AI and developer-tooling projects for EdTech platforms serving **40+ million users**. **TOEFL iBT: 5/6 (100/120 equivalent).**

Education

Amirkabir University of Technology (Tehran Polytechnic) — Tehran, Iran
B.Sc. in Computer Science — *Sep 2022 – Jul 2026*
- **GPA:** 19.55/20 (Equivalent to 4.0/4.0) — **Rank:** 1st out of 72 students

Allameh Helli 8 High School (NODET/SAMPAD) — Tehran, Iran
Diploma in Experimental Sciences — *Sep 2017 – Jun 2021*
- **GPA:** 19.23/20 (Top-ranked talented student school)

Research Experience

Submitted Manuscript: Gradient-Noise-Scale-Aware Adaptive Sampling — Feb 2026 – Present
for Communication-Efficient Vertical Federated Logistic Regression — *Amirkabir University*
- Co-authored a submitted paper proposing a gradient-noise-scale-aware adaptive sampling controller for communication-efficient vertical federated logistic regression.
- Implemented and analyzed **950 runs** across **10 datasets** and 5 seeds; the reference setting reduced sample-party communication by **90.89%** at macro ROC-AUC 0.729.

Bachelor's Thesis: Deep Learning for Knowledge Tracing — Sep 2025 – Jun 2026
Supervised Research Project — *Amirkabir University*
- Designed and evaluated transformer- and graph-based Knowledge Tracing models (DKT, AKT, GKT) on ASSISTments and EdNet for personalized learning prediction.
- Investigated attention mechanisms for interpretable modeling of student knowledge states.

AI R&D Engineer — Applied Research — Sep 2024 – Present
Rastanovin Raya (Ministry of Education Contractor, 200+ employees) — *Tehran, Iran*
- Conducting applied research on production-grade AI/ML systems powering educational platforms with **40+ million users**.
- Developing LLM-based automation pipelines reducing manual document processing time by 60%.
- Building intelligent assistant systems integrating RAG architectures with domain-specific knowledge bases for large-scale EdTech applications.

Teaching Experience

Teaching Assistant — 10+ Undergraduate & Graduate Courses — Jan 2023 – Jul 2026
Amirkabir University of Technology — *Tehran, Iran*
- Courses: Data Structures & Algorithms, Artificial Intelligence, Numerical Linear Algebra, Operating Systems, Foundations of Mathematics, Basic Programming, Logic, Principles of Computer Systems, Computer Simulation, Network Security, Introductory Probability & Statistics, Image Processing, and Data Mining.
- **Head TA for Advanced Artificial Intelligence** (M.Sc. course): Led TA team for 40+ graduate students.
- Designed 15+ programming assignments and projects; mentored 200+ students across multiple semesters.

Teaching Assistant — Operating Systems — Sep 2025 – Jul 2026
University of Tehran — *Tehran, Iran*
- Provided hands-on lab support for process management, memory allocation, and file systems projects.
- Mentored 40+ students on systems programming assignments in C.

Work Experience

AI Developer & Team Lead (Part-time) — Sep 2024 – Present
Rastanovin Raya — Ministry of Education Contractor (200+ employees) — *Tehran, Iran*
- Developing AI solutions for Iran's largest EdTech platform serving **40+ million users** (students, teachers, administrators) through Ministry of Education contracts.
- Led a **team of 2 developers** to build **Vibe Coding** features for the company's Cloud Development Environment (CDE) — an internal platform handling CI/CD, code hosting, and full-stack development workflows.
- The CDE New Style demo version is currently used by **30+ developers** building complete applications (frontend, backend, database) from scratch to production deployment.
- Built LLM-based automation pipelines for intelligent code assistance, document processing, and developer productivity tools.

- Designed ML model pipelines achieving 92% accuracy on structured document parsing at scale.

Frontend Developer Intern Jul 2023 – Sep 2023
Rastanovin Raya *Tehran, Iran*
- Developed responsive web interfaces using React and JavaScript for internal enterprise tools.
- Implemented RESTful API integrations and improved UI/UX based on user feedback.

Selected Projects

Cloud Development Environment (CDE) — Vibe Coding Features | *Python, JS, LLM, Docker* Oct 2024 – Present
- Led development of AI-powered coding features for enterprise cloud IDE serving 30+ active developers.
- Implemented context-aware code generation, intelligent autocomplete, and automated code review using LLMs.
- Integrated with existing CI/CD pipelines enabling end-to-end development workflows (frontend, backend, database).

LLM Code Assistance Tool | *Python, JavaScript, LangChain, Git* Oct 2024 – Present
- Developed a Cursor-like AI coding assistant with context-aware prompt generation and automatic code merging.
- Implemented intelligent diff-based merge algorithm achieving 95%+ successful code integrations.

RAG System | *Python, Docker, LangChain, React* Mar 2025
- Built production-ready Retrieval-Augmented Generation pipeline indexing 10,000+ documents with vector embeddings.
- Integrated fine-tuned LLMs for domain-specific Q&A; Dockerized for scalable cloud deployment.

Exam Paper Segmentation & Blank Detection | *Python, OpenCV* Aug 2024
- Designed computer vision system for Ministry of Education automating exam paper processing.
- Achieved 96% accuracy in question boundary detection and blank answer identification across 5,000+ pages.

Urban Bridge Failure Detection System | *Python, PyTorch, OpenCV, YOLO* Feb 2024 – Present
- Industrial collaboration with Tehran Urban Planning and Research Center (TUPRC) on AI-driven infrastructure inspection.
- Developed deep learning pipeline detecting structural defects (cracks, corrosion, deformation) from bridge inspection images.

Sabad — E-Commerce Platform | *React, Node.js, PostgreSQL* Jul 2023 – Sep 2023
- Developed full-stack shopping platform with user authentication, product catalog, and order management.

Fraud & ADHD Detection Models | *Python, scikit-learn, pandas* Jun 2024
- Built classification pipelines for fraud detection and ADHD prediction tasks.

Awards & Certifications

- **Certificate of Achievement — MLx Representation Learning & Generative AI 2026**, Oxford Machine Learning School (OxML), Jul 2026 — completed an intensive 18-hour programme and associated assessment.
- **Top 1%** — National University Entrance Exam (Konkur) for B.Sc. Admission
- **Qualified Candidate** — Passed 1st Round of National Mathematics Olympiad (2 consecutive years)
- **Top-Ranked Participant** — Quera Fito Hackathon (National Programming Competition)
- **Top Performer in Artificial Intelligence** — 2nd National Technology Olympiad
- **Top Performer in Frontend Development** — 2nd National Technology Olympiad
- Algorithm Arena Certificate — Scrum/Agile Certification

University Activities

Scientific Association Member & Event Staff May 2024 – Jul 2026
Computer Science Department *Amirkabir University*
- Organized technical workshops on AI/ML topics with 100+ student attendees.
- Served as staff coordinator for university programming competitions and academic events.
- Mentored junior students on academic projects and career guidance.

Technical Skills

Programming Languages: Python, C/C++, JavaScript, TypeScript, SQL (PostgreSQL), R, LaTeX
AI/ML Areas: LLMs, RAG, Federated Learning, Knowledge Tracing, Computer Vision, Image Processing, Representation Learning, Generative AI
AI/ML Libraries: PyTorch, TensorFlow, scikit-learn, LangChain, Hugging Face Transformers, OpenCV
Frameworks & Tools: React, Node.js, Docker, Git, CI/CD, REST APIs
Data & Modeling: NumPy, pandas, Matplotlib, Jupyter, Data Mining, Probability & Statistics, Computer Simulation
Systems & Development: Linux/Unix, Operating Systems, Network Security, CI/CD, VS Code, PyCharm, CLion

Languages

Persian: Native — **English:** Fluent (Academic & Professional Proficiency) — **TOEFL iBT:** 5/6 (100/120 equivalent)